NegBioDB: A Multi-Domain Database and Benchmark for Experimentally Confirmed Negative Results in Biomedicine

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Abstract

Publication bias ensures that negative experimental results—inactive compounds, failed clinical trials, confirmed non-interactions—remain largely unreported, creating systematic blind spots in biomedical AI. Machine learning benchmarks treat untested pairs as negatives, while large language models hallucinate evidence for experiments never conducted. We introduce **NegBioDB**, the first multi-domain database of experimentally confirmed negative results, aggregating 32.9 million entries from 12 data sources across three domains: drug–target interaction (DTI; 30.5M), clinical trial failure (CT; 133K), and protein–protein interaction (PPI; 2.2M), organized by four confidence tiers. We pair this resource with **NegBioBench**, a dual ML+LLM benchmark comprising 421 experiments (180 ML, 241 LLM) across four evaluation levels of increasing cognitive demand. Our experiments reveal three findings: (1) control negatives inflate DTI model performance by +0.112 LogAUC, with model-dependent effects in PPI and CT; (2) cold-entity splits expose universal generalization failures, including a sequence-based PPI model dropping to AUROC = 0.41 (below random); and (3) the *opacity gradient*—LLM discrimination between tested-negative and untested pairs correlates with data accessibility in training corpora (DTI MCC 0.18 $\rightarrow$ PPI 0.44 $\rightarrow$ CT 0.56), not biological reasoning capability. Temporal contamination analysis confirms that PPI performance reflects memorization rather than understanding. All five evaluated LLMs exhibit 100% evidence hallucination rates across all domains. NegBioDB and NegBioBench are released under CC BY-SA 4.0 with Croissant metadata at https://github.com/jang1563/NegBioDB.

1 Introduction

Approximately 90% of experiments produce null or inconclusive results, yet publication bias ensures the vast majority remain unreported [Fanelli, 2012, Mlinaric et al., 2017]. This creates a systematic blind spot in biomedical AI: machine learning benchmarks treat untested compound–target pairs as negatives [Huang et al., 2021, Mysinger et al., 2012], an assumption that inflates reported performance by up to +0.112 LogAUC and produces models incapable of distinguishing genuinely tested inactive pairs from untested ones (MCC $\leq$ 0.18). Meanwhile, large language models confidently generate hallucinated evidence for 100% of queried negative results, regardless of whether the experiment was ever conducted.

The consequences span three biomedical domains where negative results carry critical information. In drug–target interaction (DTI), less than 1% of compound–target space has been experimentally tested [Gaulton et al., 2017], yet benchmarks like TDC [Huang et al., 2021] and DUD-E [Mysinger

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Table 1: NegBioDB database overview across three biomedical domains.

	DTI	**CT**	**PPI**	**Total**
Negative results	30.5M	132,925	2.23M	32.9M
Key entities	919K / 3.7K	177K / 56K	18.4K	—
	(cpd / tgt)	(interv / cond)	(proteins)	
Data sources	4	4	4	12
Confidence tiers	3	4	3	4
DB size	13.2 GB	0.5 GB	0.8 GB	14.6 GB
ML benchmark runs	18	108	54	180
LLM benchmark runs	81	80	80	241

et al., 2012] assume all untested pairs are negative. In clinical trials (CT), failed trials vastly outnumber successes but lack structured representation—the AACT database [Tasneem et al., 2012] contains 216K trials with rich failure metadata that no existing benchmark leverages. In protein–protein interaction (PPI), systematic screens such as HuRI [Luck et al., 2020] produce millions of confirmed non-interactions that benchmarks ignore in favor of random negative sampling.

No existing benchmark evaluates negative result understanding. ChemBench [Mirza et al., 2024] tests chemical property reasoning; Mol-Instructions [Fang et al., 2024] evaluates molecular understanding; LAB-Bench [Laurent et al., 2024] measures laboratory skills; MedQA [Jin et al., 2021] tests clinical knowledge—but none specifically address the challenge of distinguishing tested from untested hypotheses, which is fundamental to the scientific process. Recent work on DTI negative evidence [Li et al., 2025] and benchmark quality [Volkov et al., 2025, Tran-Nguyen et al., 2020] highlights the growing recognition of this gap, but remains limited to single domains.

We address this gap with four contributions:

1. **NegBioDB**: the first multi-domain database of experimentally confirmed negative results, aggregating 32.9M entries from 12 data sources across DTI, CT, and PPI with four confidence tiers (Section 2). Released under CC BY-SA 4.0 with Croissant metadata.
2. **NegBioBench**: a dual ML+LLM benchmark with 4 evaluation levels across 3 domains, totaling 421 experiments (180 ML + 241 LLM) that systematically test both predictive models and language models on negative evidence understanding (Section 3).
3. **Negative source inflation**: we demonstrate that control negatives inflate DTI model LogAUC by +0.112 and show model-dependent effects in PPI (-0.11 to $+0.05$ LogAUC) and CT (-0.16 to -0.24 AUROC), challenging the universal assumption that negative source does not matter (Section 4.1).
4. **The opacity gradient**: L4 discrimination performance correlates with data accessibility in LLM training corpora (DTI MCC 0.18 $\rightarrow$ PPI 0.44 $\rightarrow$ CT 0.56), with PPI contamination confirmed as genuine memorization via temporal stratification and protein degree analysis (Section 4.4).

2 NegBioDB: A Database of Negative Results

NegBioDB is a multi-domain database of experimentally confirmed negative results in biomedicine, aggregating 32.9M entries from 12 data sources across three domains. We first describe the common design principles, then detail each domain.

2.1 Design Principles

All three domains share a common abstraction layer: each record encodes a *hypothesis* (e.g., "compound X inhibits target Y"), *experimental evidence* (assay type, method, publication), an *outcome* (inactive, failed, non-interacting), and a *confidence tier* reflecting evidence quality. The four-tier system is: **Gold**—systematic screens or multiple independent confirmations (e.g., DAVIS kinase panel, HuRI Y2H screen); **Silver**—single quantitative measurement or statistical evidence (e.g., $p > 0.05$ from clinical trial, ML-derived from co-purification data); **Bronze**—computationally derived or NLP-detected (e.g., STRING zero-score pairs, NLP-classified trial terminations); **Copper**—label-only annotations without detailed evidence. Table 1 summarizes the database scope.

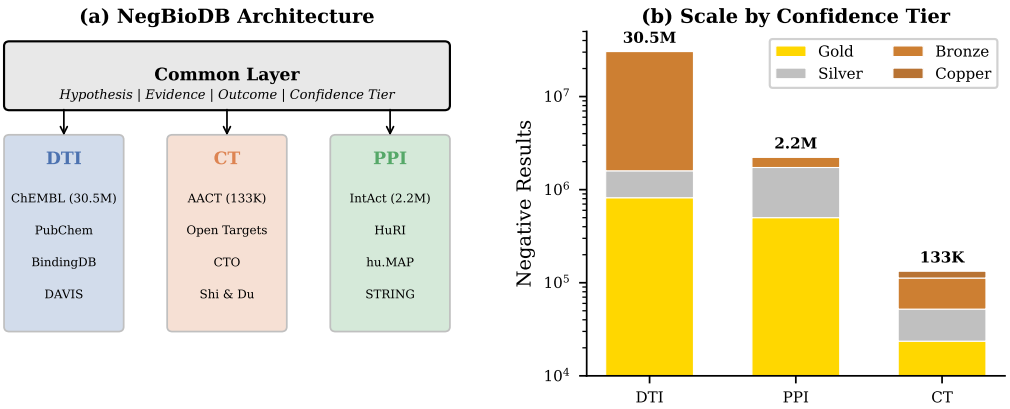

Figure 1: NegBioDB overview. **(a)** Architecture showing three domains unified by a common abstraction layer with four confidence tiers. Each domain integrates four data sources. **(b)** Scale of negative results by domain and confidence tier (log scale). DTI dominates in volume (30.5M), while CT and PPI contribute qualitatively distinct evidence types.

2.2 Three Domains

Drug–Target Interaction (DTI). We aggregate inactive compound–target pairs from four sources: ChEMBL [Gaulton et al., 2017] bioactivity records with pChEMBL < 5 (i.e., $IC_{50} > 10\,\mu$M); PubChem [Kim et al., 2023] confirmatory inactives from dose-response screens; BindingDB [Gilson et al., 2016] entries with $K_d > 10\,\mu$M; and the full DAVIS kinase selectivity matrix [Davis et al., 2011], where untested pairs are excluded. This yields 30.5M negative results across 919K compounds and 3,694 targets—three orders of magnitude larger than standard DTI benchmarks that rely on assumed negatives [Huang et al., 2021, Mysinger et al., 2012].

Clinical Trial Failure (CT). We process 216,987 trials from the AACT database [Tasneem et al., 2012] through a three-tier failure detection pipeline: (i) NLP classification of termination reasons into 7 failure categories (bronze tier); (ii) statistical evidence extraction from outcome measures where $p > 0.05$ indicates non-superiority (silver/gold tiers); and (iii) integration of the Clinical Trial Outcome dataset [Siah et al., 2021] for label-only records (copper tier). Drug names are resolved to ChEMBL identifiers through a four-step cascade (exact match, PubChem API, fuzzy matching with Jaro–Winkler > 0.90, manual curation), achieving 20.6% resolution with SMILES structures. The pipeline identifies 132,925 failure results with 8 failure categories: safety, efficacy, enrollment, strategic, regulatory, design, pharmacokinetic, and other.

Protein–Protein Interaction (PPI). We compile confirmed non-interactions from four sources spanning different evidence types: IntAct [Orchard et al., 2014] curated non-interactions from co-immunoprecipitation and two-hybrid assays (779 gold/silver pairs); HuRI [Luck et al., 2020] systematic yeast two-hybrid screen negatives sampled from 39.9M candidates via reservoir sampling (500K gold pairs); hu.MAP [Drew et al., 2021] ML-derived non-interactions from co-purification mass spectrometry (1.23M silver pairs); and STRING [Szklarczyk et al., 2023] zero-score pairs between well-studied proteins (500K bronze pairs). After cross-source aggregation, NegBioDB contains 2.23M unique negative PPI pairs across 18,412 human proteins with UniProt-validated identifiers and sequences (99.6% coverage).

3 NegBioBench: Evaluation Framework

NegBioBench is a dual ML+LLM benchmark designed to evaluate both predictive models and language models on negative biological evidence. We describe the two evaluation tracks and our methodology for measuring genuine understanding versus memorization.

3.1 ML Track

The ML track evaluates whether predictive models can distinguish experimentally confirmed negatives from positives, and whether standard evaluation practices inflate reported performance.

Tasks. We define two task types: *M1* (binary classification: negative vs. positive) across all three domains, and *M2* (7-way failure category prediction) for CT only. Positive examples come from established sources: DAVIS actives for DTI, CTO successful trials [Siah et al., 2021] for CT, and HuRI positive interactions [Luck et al., 2020] for PPI.

Splitting strategies. We implement domain-appropriate cold splits to test generalization: cold_drug and cold_target (DTI), cold_drug and cold_condition (CT), cold_protein and cold_both via METIS graph partitioning [Karypis and Kumar, 1998] (PPI), plus random, temporal, scaffold [Bemis and Murcko, 1996], and degree-balanced (DDB) [Zheng and Wu, 2020] splits where applicable.

Control negatives. To measure the effect of negative source on model performance (*Experiment 1*), we train identical models on NegBioDB negatives versus two control sets: uniform random pairs and degree-matched random pairs. This directly tests whether curated negatives carry different signal than assumed negatives.

Models. Three architectures per domain: DeepDTA [Öztürk et al., 2018], GraphDTA [Nguyen et al., 2021], and DrugBAN [Bai et al., 2023] for DTI; XGBoost [Chen and Guestrin, 2016], MLP, and GNN for CT; SiameseCNN, PIPR [Chen et al., 2019], and MLPFeatures for PPI. Metrics include AUROC, $\text{LogAUC}_{[0.001,0.1]}$ [Mysinger and Shoichet, 2010], AUPRC, and MCC [Matthews, 1975].

3.2 LLM Track

The LLM track evaluates language models across four levels of increasing cognitive demand:

L1 (Multiple Choice). Classification of negative evidence into domain-specific categories (4-way for DTI/PPI, 5-way for CT). Tests whether LLMs can recognize evidence types from textual descriptions.

L2 (Extraction). Structured JSON extraction of key fields from evidence text (compound/target identifiers, assay types, p-values). Tests whether LLMs can parse scientific evidence into machine-readable formats.

L3 (Reasoning). Open-ended scientific reasoning about why a negative result was observed and its implications. Evaluated by an LLM-as-judge on four dimensions: accuracy, completeness, reasoning quality, and specificity.

L4 (Discrimination). Binary classification of whether a given entity pair has been *experimentally tested and found inactive* versus *never tested*. This is the critical level: it tests whether LLMs possess genuine understanding of negative results or merely recall information from training data.

Models and configurations. Five models span the capability spectrum: Llama-3.3-70B [Dubey et al., 2024], Qwen2.5-32B [Yang et al., 2024], GPT-4o-mini [OpenAI, 2024], Gemini-2.5-Flash [Google DeepMind, 2025], and Claude Haiku-4.5 [Anthropic, 2025]. Each is evaluated in zero-shot and three few-shot configurations (different random example sets), yielding 4 configurations per model–level pair.

3.3 Evaluation Methodology

NegBioBench makes three methodological contributions relevant to the evaluation landscape:

L4 as a contamination probe. L4 discrimination performance reveals whether models have memorized negative results from training data. We incorporate temporal stratification (pre-2015 vs. post-2020 publication dates) to detect contamination: a performance gap exceeding 0.15 indicates likely memorization rather than reasoning [Sainz et al., 2024].

Cross-domain comparison. By applying identical evaluation levels across three domains with different data accessibility profiles, we can isolate the effect of training data composition on LLM performance—independent of biological task difficulty.

Table 2: ML results summary: best model per domain across key splits and negative source inflation. Cold-X/Y denote the domain-specific cold entity splits (drug/target for DTI, drug/condition for CT, protein/both for PPI). Δ reports the change in primary metric when switching from NegBioDB to degree-matched negatives.

Domain	Model	Random	Cold-X	Cold-Y	DDB	Neg. Inflation
DTI	GraphDTA	.997	.997	.863	.997	+0.124 LogAUC
DTI	DrugBAN	.997	.997	.760	.997	+0.125 LogAUC
CT-M1	XGBoost	1.00	1.00	1.00	—	−0.16 AUROC
CT-M2	XGBoost	.51 mF1	.41	.34	—	—
PPI (seq)	PIPR	.964	.859	**.409**	.964	+0.03–0.05 LogAUC
PPI (feat)	MLPFeat	.962	.931	**.950**	.961	**−0.03 to −0.11 LogAUC**

Anti-contamination design. L1–L3 use paraphrased evidence text to reduce verbatim memorization advantage. L4 temporal holdouts ensure post-training-cutoff examples are included. All hallucinated evidence citations are tracked to measure confabulation rates.

4 Experiments and Results

We organize 421 experiments (180 ML + 241 LLM) into four cross-domain findings. DTI uses a single seed; CT and PPI use 3 seeds (we report mean $\pm$ std). Full per-run tables appear in Appendices B–C.

4.1 Do Curated Negatives Differ from Controls?

We train identical models on NegBioDB negatives versus uniform random and degree-matched control negatives, holding all other variables constant (Table 2).

DTI. Degree-matched negatives inflate LogAUC by +0.112 on average across all three models (e.g., GraphDTA: 0.843 $\rightarrow$ 0.967). This confirms that assumed-negative benchmarks systematically overestimate model performance. Uniform random controls show negligible average inflation ($<$+0.01 LogAUC), indicating that degree matching specifically creates an especially easy discrimination task.

CT. The pattern reverses in direction: NegBioDB negatives (clinical failures) are trivially separable from CTO successes (AUROC = 1.0), while control negatives are harder. Degree-matched controls reduce AUROC by 0.16–0.24 across models (e.g., GNN: 1.0 $\rightarrow$ 0.76). This is expected—genuine clinical failures carry rich pharmacological features absent from random drug–condition pairs.

PPI. The effect is *model-dependent*—a novel finding. Sequence-based models (SiameseCNN, PIPR) show +0.03–0.05 LogAUC inflation with control negatives, consistent with DTI. However, MLPFeatures shows *reversed* inflation: NegBioDB negatives are harder than controls (-0.03 to -0.11 LogAUC), because hand-crafted features (protein degree, subcellular localization) capture the same structural signal that curated negatives encode.

Cross-domain insight. Curated negatives carry systematically different signal than controls in all three domains. The *direction* of inflation depends on whether the model architecture captures the same features that distinguish curated from random negatives.

4.2 Does Cold Splitting Expose Generalization Failures?

Cold splits remove all instances of a specific entity from the training set, testing whether models generalize to unseen drugs, targets, proteins, or conditions (Figure 2).

DTI. Cold-target splitting is catastrophic: LogAUC collapses from 0.83 to 0.15–0.33 across all three models, while AUROC misleadingly remains 0.76–0.89. DrugBAN suffers most severely (LogAUC = 0.151, AUROC = 0.760). Cold-compound splitting has minimal effect, indicating models memorize target-specific patterns.

CT. M1 binary classification remains trivially solvable even under cold splits (AUROC = 1.0 for cold_drug and cold_condition). However, M2 7-way category prediction reveals severe failures:

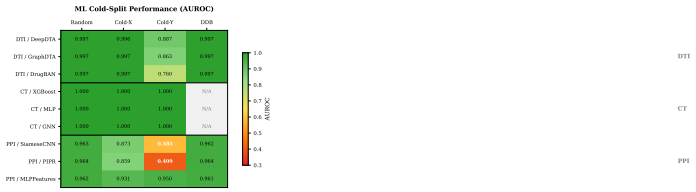

Figure 2: ML cold-split performance across domains (AUROC). Red cells indicate catastrophic failure (< 0.7). CT M1 is trivially separable across all splits; cold-both PPI reveals a 0.54 gap between sequence (PIPR: 0.409) and feature-based (MLPFeatures: 0.950) architectures. N/A: split not applicable.

scaffold and temporal splits collapse macro-F1 to 0.19 across all models, approaching the random baseline of $1/7 \approx 0.14$.

PPI. Cold-both splitting (unseen proteins on both sides, via METIS partitioning) produces a model-dependent catastrophe. PIPR drops to AUROC = 0.409—*below random*—while MLPFeatures remains robust at 0.950. This 0.54 AUROC gap between sequence-based and feature-based architectures is the largest we observe. SiameseCNN falls in between (0.585).

Cross-domain pattern. Cold-split catastrophe is *universal* across domains but *model-dependent* within each domain. Sequence-based and attention models memorize training entities; feature-based models generalize. Notably, DDB $\approx$ random in all three domains, suggesting degree-balanced splitting does not add meaningful difficulty beyond random assignment.

4.3 Can LLMs Reason About Negative Biological Evidence?

We evaluate five LLMs across four evaluation levels in each domain (Table 3).

L1 (Multiple Choice). Performance varies dramatically by domain. PPI is trivially solvable: all models achieve $\geq$0.999 accuracy with 3-shot prompting (evidence text makes category self-evident). CT is the hardest domain: the best model (Gemini) reaches only 0.667 accuracy on 5-way classification, well below the 0.80+ levels seen in DTI and PPI. DTI falls between (0.65–1.0 with 3-shot). Difficulty correlates with label discriminability in evidence text.

L2 (Extraction). PPI extraction is near-perfect (entity F1 $\geq$ 0.95, count accuracy 1.0) because protein names are unambiguous identifiers. CT is moderately difficult (field F1: 0.48–0.81) due to complex clinical evidence with multiple p-values and outcomes. DTI L2 was not evaluated due to the annotation cost of gold-standard evidence text.

L4 (Discrimination). This is the critical level testing genuine understanding. The results reveal a striking *gradient* across domains:

- **DTI**: MCC $\leq$ 0.184 — near random. LLMs cannot distinguish tested-inactive from untested compound–target pairs.
- **PPI**: MCC 0.33–0.44 — moderate. Some discrimination ability, but contamination analysis (Section 4.4) reveals this is largely memorization.
- **CT**: MCC 0.48–0.56 — meaningful. Gemini achieves 0.563, the highest discrimination across all domains.

Evidence hallucination. Across all domains, models, and levels, 100% of generated evidence citations are hallucinated—models never cite real PMIDs or DOIs. This universal confabulation rate persists even when models make correct predictions.

L3 reasoning results are deferred to Appendix D due to a ceiling effect: the GPT-4o-mini judge assigns 4.4–5.0/5.0 for CT, confounding cross-domain comparison.

4.4 The Opacity Gradient

The L4 discrimination results in Section 4.3 reveal a pattern we term *the opacity gradient*: LLM performance correlates not with biological task difficulty, but with the accessibility of domain data in LLM training corpora (Figure 3).

Table 3: LLM cross-domain results (best configuration per domain). L3 omitted from main text (see Appendix D). †Contamination gap: pre-2015 minus post-2020 accuracy.

Level	Metric	DTI	CT	PPI
L1 MCQ	Accuracy	Llama 0.991	Gemini 0.667	Llama 1.000
L2 Extract	Field F1	—	Qwen 0.81	Haiku 1.000
L4 Discrim	**MCC**	**Llama 0.184**	**Gemini 0.563**	**Llama 0.441**
L4 Contam.	Gap†	<0.15	<0.15	**0.36–0.61**
Hallucination	Rate	100%	100%	100%

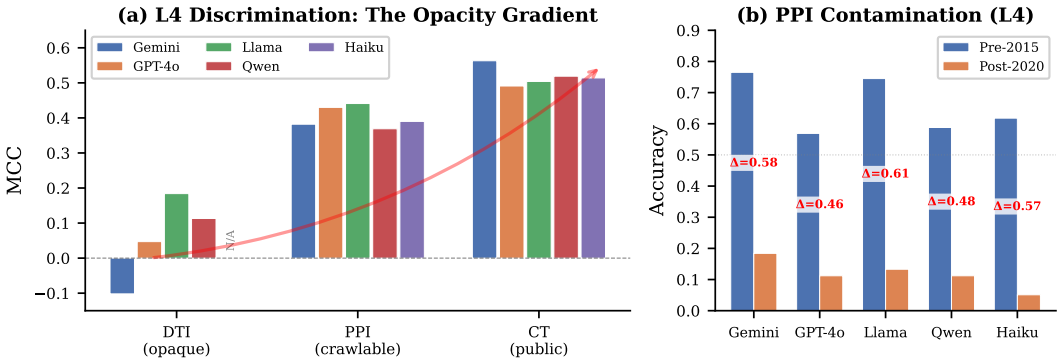

Figure 3: The opacity gradient. **(a)** L4 discrimination (MCC) across domains for four common models (+Haiku for CT/PPI). Performance increases with data accessibility: DTI (opaque databases) → PPI (crawlable) → CT (public). Dashed line: MCC = 0. **(b)** PPI contamination: pre-2015 vs. post-2020 accuracy. All models show >0.35 gaps, confirming memorization. Δ values in red.

DTI data is opaque. ChEMBL bioactivity tables store compound–target interactions in structured databases behind query interfaces. Individual IC_{50} values are unlikely to appear in web crawls used for LLM pretraining. Result: MCC $\leq$ 0.18 (near random).

PPI data is crawlable. IntAct and STRING expose protein interaction data through publicly crawlable web pages and bulk downloads frequently indexed by search engines. Result: MCC 0.33–0.44 (moderate), but contamination analysis reveals this is memorization.

CT data is public. ClinicalTrials.gov trial records are heavily discussed in news articles, regulatory filings, investor reports, and medical literature. Result: MCC 0.48–0.56 (meaningful discrimination).

PPI contamination confirmed. All five models show large temporal accuracy gaps (0.36–0.61) between pre-2015 and post-2020 interaction pairs, far exceeding the 0.15 contamination threshold [Balloccu et al., 2024]. Pre-2015 pairs—available in training data—are classified with 40–79% accuracy, while post-2020 pairs drop to 2–24%. To rule out protein popularity as a confound, we stratify by protein degree (Appendix E): contamination persists in both high-degree and low-degree protein pairs (gaps 0.33–0.58), confirming genuine memorization rather than popularity bias.

No contamination is detected for DTI (MCC too low to measure temporal effects) or CT (gaps <0.15).

Implication. L4 performance reflects training data composition, not biological reasoning capability. LLMs are fundamentally *failure-blind*—they cannot distinguish tested-negative from untested pairs without prior memorization. This finding has direct implications for LLM deployment in drug discovery and clinical trial design, where hallucinated confidence in untested hypotheses could misdirect research effort.

5 Discussion and Conclusion

The opacity gradient and its implications. The opacity gradient implies that LLM benchmarks must account for data accessibility in training corpora, not just task difficulty. The L4 progression—

DTI (MCC $\leq$ 0.18) $\rightarrow$ PPI (0.44) $\rightarrow$ CT (0.56)—does not reflect increasing biological reasoning capability. Rather, it mirrors the degree to which each domain's data appears in web crawls: ChEMBL bioactivity tables are locked behind database queries (opaque), IntAct/STRING protein interaction data is publicly crawlable (memorizable), and ClinicalTrials.gov records are extensively discussed in news, regulatory filings, and medical literature (public). Models do not "understand" negative results—they recall them. This distinction is critical for responsible LLM deployment in drug discovery and clinical trial design.

Curated negatives as a feature. NegBioDB negatives being trivially separable from positives by ML models (AUROC = 1.0 on random splits) is a feature, not a bug: it confirms that experimentally confirmed negatives encode genuine biological signal absent from random pairs. The value of NegBioDB for ML emerges in cold and temporal splits, where models must generalize to unseen entities—and where we observe catastrophic failures (PIPR cold_both AUROC = 0.409) that random negatives would not reveal.

Universal evidence hallucination. The 100% hallucination rate for evidence citations across all 241 LLM runs is a safety concern. Even when models correctly classify negative results, they fabricate supporting evidence (PMIDs, DOIs, author names). This confabulation pattern persists across all five models, all four levels, and all three domains, suggesting it is a fundamental limitation of current LLMs rather than a model-specific issue.

Limitations. DTI ML baselines use a single seed (CT/PPI use 3 seeds). CT drug resolution covers only 20.6% of interventions due to non-standard drug naming in trial records. PPI L1/L2 tasks are trivially solvable with few-shot prompting, limiting their discriminative value. The L3 reasoning evaluation suffers from a judge ceiling effect in CT (Appendix D). Contamination analysis is conclusive only for PPI; DTI L4 performance is too low to detect temporal patterns. The database was developed by a single author, mitigated by 800+ automated tests and a comprehensive reproducibility pipeline.

Conclusion. NegBioDB provides the first multi-domain resource for experimentally confirmed negative results in biomedicine (32.9M entries, CC BY-SA 4.0, Croissant metadata). NegBioBench reveals that curated negatives carry systematically different signal than controls, cold splits expose universal generalization failures, and LLMs are fundamentally failure-blind—unable to distinguish tested from untested pairs without training data memorization. We release all databases, benchmarks, and code to support future work on negative result understanding, including community contribution tools, additional domains (e.g., gene function), and a public leaderboard.

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A Datasheet for Datasets

Following the framework of Gebru et al. [2021], we provide a datasheet for NegBioDB.

A.1 Motivation

For what purpose was the dataset created? NegBioDB was created to address the absence of curated negative results in biomedical AI benchmarks. Existing benchmarks treat untested entity pairs as negatives, an assumption that inflates model performance and prevents evaluation of genuine negative result understanding.

Who created the dataset and on behalf of which entity? The dataset was created by a single researcher at Weill Cornell Medicine as part of a doctoral research project.

Who funded the creation of the dataset? No external funding was received for this project.

A.2 Composition

What do the instances that comprise the dataset represent? Each instance represents an experimentally confirmed negative result: a biological hypothesis that was tested and found to be unsupported. Specifically:

- **DTI**: A compound–target pair tested for binding activity and found inactive (e.g., $IC_{50} > 10\mu$M).
- **CT**: A drug–condition pair tested in a clinical trial that failed to meet its primary endpoint.
- **PPI**: A protein–protein pair tested for physical interaction and found non-interacting.

How many instances are there in total? 32.9 million negative results: 30.5M (DTI), 132,925 (CT), and 2.23M (PPI).

Does the dataset contain all possible instances or is it a sample? It is a curated sample from 12 source databases. DTI includes all ChEMBL pchembl<5 records, PubChem confirmatory inactives, BindingDB Kd>10μM, and DAVIS matrix inactives. CT includes all AACT trials classified as clinical failures plus CTO copper-tier records. PPI includes IntAct curated non-interactions, HuRI screen negatives (sampled from 39.9M), hu.MAP ML-derived negatives, and STRING zero-score pairs (sampled from >100M).

What data does each instance consist of? Entity identifiers (ChEMBL IDs, UniProt accessions, NCT IDs), experimental metadata (assay type, detection method, p-values), outcome measures (activity values, effect sizes), confidence tier assignment, and provenance information (source database, extraction method, publication year).

Is there a label or target associated with each instance? Yes. All instances are labeled as negative results with a four-tier confidence classification: gold (systematic screens, multiple confirmations), silver (ML-derived or p-value based), bronze (computational or NLP-detected), copper (label-only, minimal evidence).

Is any information missing from individual instances? Activity values are missing for some DTI records (especially PubChem inactives that report only active/inactive). Clinical trial p-values are available only for gold/silver tier CT records. PPI records from STRING lack experimental metadata.

Are there any errors, sources of noise, or redundancies? Potential errors include: NLP misclassification of CT failure categories (bronze tier), false negatives in HuRI Y2H screens (estimated 20–40% false negative rate for Y2H), and activity value discrepancies across sources for DTI. Deduplication indexes prevent exact duplicates but near-duplicate records from different sources are preserved as multi-source evidence.

Is the dataset self-contained? Yes. The SQLite databases contain all necessary data. Chemical structures (SMILES), protein sequences, and trial metadata are stored directly. External identifiers enable linkage to source databases for additional context.

A.3 Collection Process

How was the data associated with each instance acquired? All data was acquired from public databases via their official APIs or bulk download services:

- ChEMBL v34 (SQL dump), PubChem (BioAssay FTP), BindingDB (servlet download), DAVIS (literature supplement)
- AACT (monthly PostgreSQL dump), CTO (GitHub release), Open Targets (API), Shi & Du 2024 (supplement)
- IntAct (PSI-MI TAB), HuRI (interactome-atlas.org), hu.MAP 3.0 (bulk download), STRING v12.0 (API)

What mechanisms or procedures were used to collect the data? Automated ETL pipelines with validation checks. Drug names in CT were resolved to ChEMBL identifiers via a 4-step cascade: exact ChEMBL match → PubChem API → fuzzy matching (Jaro-Winkler > 0.90) → manual CSV. CT failure categories were assigned via 3-tier detection: NLP keyword matching → p-value extraction → CTO labels.

Who was involved in the data collection process? A single researcher performed all data collection, processing, and validation. 800+ automated tests verify pipeline correctness.

Over what timeframe was the data collected? Data collection occurred January–March 2026. Source databases span different time periods: ChEMBL v34 (through 2024), AACT (through February 2026), IntAct (through 2024), HuRI (2020 publication).

A.4 Preprocessing, Cleaning, Labeling

Was any preprocessing/cleaning/labeling of the data done?

- **DTI**: SMILES canonicalization via RDKit, InChIKey generation, pchembl value computation, activity value unit normalization. Confidence tier assignment based on source and assay type.
- **CT**: Three-tier failure classification (NLP/p-value/CTO), drug name resolution to ChEMBL, failure category assignment (8 categories). Tier upgrades applied (bronze+p-value→silver).
- **PPI**: UniProt accession validation, ENSG version suffix stripping, canonical pair ordering (protein1_id $<$ protein2_id), reservoir sampling for HuRI and STRING.

Was the "raw" data saved in addition to the preprocessed/cleaned/labeled data? Source database downloads are archived but not distributed due to size (AACT: 2.23 GB, ChEMBL: several GB). The processed SQLite databases and Parquet exports are the distributed artifacts.

A.5 Uses

Has the dataset been used for any tasks already? Yes, for the NegBioBench benchmark described in this paper: 180 ML experiments and 241 LLM experiments across three domains.

What (other) tasks could the dataset be used for? Drug repurposing (negative results constrain hypothesis space), clinical trial design (learning from prior failures), protein interaction network refinement, negative-aware training for DTI/PPI prediction models, and LLM evaluation for scientific reasoning.

Is there anything about the composition of the dataset or the way it was collected that might impact future uses? The CC BY-SA 4.0 license (required by ChEMBL's CC BY-SA 3.0 viral clause) means derivative works must maintain the same license. DTI bronze tier (94.6% of records) has lower confidence than gold/silver tiers. CT drug resolution achieved only 20.6% ChEMBL coverage, limiting chemical feature availability.

A.6 Distribution

How will the dataset be distributed? Via HuggingFace Datasets Hub (primary), GitHub repository (code + small exports), and Zenodo (archival DOI).

When will the dataset be released? Upon paper acceptance or preprint posting.

Will the dataset be distributed under a copyright or IP license? CC BY-SA 4.0 International.

Have any third parties imposed IP-based or other restrictions? ChEMBL's CC BY-SA 3.0 license requires share-alike, which propagates to the full dataset. All other sources use permissive licenses (public domain, Apache 2.0, MIT, CC BY 4.0).

A.7 Maintenance

Who will be supporting/hosting/maintaining the dataset? The first author, with institutional support from Weill Cornell Medicine. HuggingFace provides persistent hosting.

How can the owner/curator/manager of the dataset be contacted? Via the GitHub repository issue tracker or the corresponding author email listed in the paper.

Will the dataset be updated? We plan annual updates incorporating new ChEMBL releases, AACT snapshots, and additional PPI databases. Version tags and checksums enable reproducible access to specific releases.

Will older versions of the dataset continue to be available? Yes, via Zenodo DOI versioning and HuggingFace dataset revisions.

If others want to contribute to the dataset, is there a mechanism? The schema includes `community_submitted` extraction method and `curator_validated` flags. A contribution platform is planned for future work.

B Complete ML Results

This appendix presents complete per-run ML results for all 180 experiments across three domains. DTI uses a single seed (42); CT and PPI use three seeds (42, 43, 44). PPI results are reported as mean $\pm$ std across seeds.

B.1 DTI ML Results (18 runs)

Table 4: DTI ML results: 3 models $\times$ 6 configurations, seed 42.

Model	Split	Negatives	LogAUC	AUPRC	MCC	AUROC
DeepDTA	random	negbiodb	0.833	0.997	0.976	0.997
DeepDTA	random	uniform_random	0.824	0.995	0.939	0.994
DeepDTA	random	degree_matched	0.919	0.998	0.980	0.998
DeepDTA	cold_compound	negbiodb	0.792	0.995	0.975	0.996
DeepDTA	cold_target	negbiodb	0.325	0.901	0.041	0.887
DeepDTA	ddb	negbiodb	0.824	0.996	0.975	0.997
GraphDTA	random	negbiodb	0.843	0.997	0.977	0.997
GraphDTA	random	uniform_random	0.888	0.996	0.947	0.996
GraphDTA	random	degree_matched	0.967	0.999	0.981	0.999
GraphDTA	cold_compound	negbiodb	0.823	0.996	0.976	0.997
GraphDTA	cold_target	negbiodb	0.241	0.871	0.098	0.863
GraphDTA	ddb	negbiodb	0.840	0.997	0.977	0.997
DrugBAN	random	negbiodb	0.830	0.996	0.975	0.997
DrugBAN	random	uniform_random	0.825	0.995	0.933	0.994
DrugBAN	random	degree_matched	0.955	0.999	0.980	0.999
DrugBAN	cold_compound	negbiodb	0.828	0.996	0.976	0.997
DrugBAN	cold_target	negbiodb	0.151	0.782	0.186	0.760
DrugBAN	ddb	negbiodb	0.828	0.996	0.975	0.997

B.2 CT-M1 Results (54 runs)

Table 5: CT-M1 binary classification results: 3 models × 6 splits × 3 seeds. Temporal split produces single-class validation sets (all negative), yielding undefined metrics (—).

Model	Split	Neg.	Seed	AUROC	AUPRC	MCC	LogAUC	F1
GNN	cold_cond	negbiodb	42	1.000	1.000	0.994	0.998	0.997
	cold_cond	negbiodb	43	1.000	1.000	0.990	0.991	0.995
	cold_cond	negbiodb	44	1.000	1.000	0.990	0.989	0.995
	cold_drug	negbiodb	42	1.000	1.000	0.993	0.999	0.997
	cold_drug	negbiodb	43	1.000	1.000	0.993	0.999	0.997
	cold_drug	negbiodb	44	1.000	1.000	0.991	0.999	0.995
	random	deg_match	42	0.724	0.724	0.393	0.170	0.636
	random	deg_match	43	0.768	0.737	0.454	0.137	0.639
	random	deg_match	44	0.781	0.741	0.473	0.132	0.657
	random	negbiodb	42	1.000	1.000	1.000	1.000	1.000
	random	negbiodb	43	1.000	1.000	1.000	1.000	1.000
	random	negbiodb	44	1.000	1.000	1.000	1.000	1.000
	random	unif_rand	42	0.891	0.914	0.609	0.305	0.841
	random	unif_rand	43	0.899	0.922	0.638	0.331	0.851
	random	unif_rand	44	0.893	0.922	0.599	0.373	0.835
	temporal	negbiodb	42	—	—	—	—	—
	temporal	negbiodb	43	—	—	—	—	—
	temporal	negbiodb	44	—	—	—	—	—
MLP	cold_cond	negbiodb	42	1.000	1.000	0.995	0.989	0.997
	cold_cond	negbiodb	43	0.999	0.999	0.990	0.980	0.995
	cold_cond	negbiodb	44	1.000	1.000	0.984	0.991	0.992
	cold_drug	negbiodb	42	1.000	0.999	0.991	0.995	0.996
	cold_drug	negbiodb	43	1.000	0.999	0.996	0.993	0.998
	cold_drug	negbiodb	44	1.000	0.999	0.996	0.997	0.998
	random	deg_match	42	0.799	0.794	0.447	0.179	0.729
	random	deg_match	43	0.803	0.802	0.462	0.195	0.728
	random	deg_match	44	0.802	0.800	0.454	0.189	0.731
	random	negbiodb	42	1.000	1.000	0.994	0.990	0.999
	random	negbiodb	43	1.000	1.000	0.994	0.993	0.999
	random	negbiodb	44	1.000	1.000	0.988	0.996	0.998
	random	unif_rand	42	0.884	0.886	0.597	0.343	0.796
	random	unif_rand	43	0.888	0.889	0.609	0.366	0.807
	random	unif_rand	44	0.884	0.888	0.592	0.387	0.786
	temporal	negbiodb	42	—	—	—	—	—
	temporal	negbiodb	43	—	—	—	—	—
	temporal	negbiodb	44	—	—	—	—	—
XGBoost	cold_cond	negbiodb	42	1.000	1.000	1.000	1.000	1.000
	cold_cond	negbiodb	43	1.000	1.000	1.000	1.000	1.000
	cold_cond	negbiodb	44	1.000	1.000	1.000	1.000	1.000
	cold_drug	negbiodb	42	1.000	1.000	0.999	1.000	1.000
	cold_drug	negbiodb	43	1.000	1.000	0.999	1.000	1.000
	cold_drug	negbiodb	44	1.000	1.000	0.999	1.000	1.000
	random	deg_match	42	0.844	0.846	0.553	0.260	0.772
	random	deg_match	43	0.844	0.846	0.553	0.260	0.772
	random	deg_match	44	0.844	0.846	0.553	0.260	0.772
	random	negbiodb	42	1.000	1.000	1.000	1.000	1.000
	random	negbiodb	43	1.000	1.000	1.000	1.000	1.000
	random	negbiodb	44	1.000	1.000	1.000	1.000	1.000
	random	unif_rand	42	0.905	0.907	0.643	0.423	0.821
	random	unif_rand	43	0.905	0.907	0.643	0.423	0.821
	random	unif_rand	44	0.905	0.907	0.643	0.423	0.821
	temporal	negbiodb	42	—	—	—	—	—
	temporal	negbiodb	43	—	—	—	—	—
	temporal	negbiodb	44	—	—	—	—	—

B.3 CT-M2 Results (54 runs)

Table 6: CT-M2 seven-way failure category prediction: 3 models $\times$ 6 splits $\times$ 3 seeds. XGBoost results are deterministic across seeds.

Model	Split	Seed	Macro-F1	Wtd-F1	MCC	Acc
GNN	cold_condition	42	0.379	0.640	0.475	0.610
	cold_condition	43	0.373	0.633	0.464	0.602
	cold_condition	44	0.377	0.629	0.460	0.596
	cold_drug	42	0.232	0.588	0.374	0.559
	cold_drug	43	0.229	0.583	0.372	0.561
	cold_drug	44	0.245	0.547	0.337	0.490
	degree_balanced	42	0.458	0.684	0.538	0.661
	degree_balanced	43	0.471	0.693	0.552	0.674
	degree_balanced	44	0.453	0.673	0.526	0.651
	random	42	0.459	0.669	0.520	0.643
	random	43	0.476	0.674	0.530	0.653
	random	44	0.469	0.674	0.529	0.657
	scaffold	42	0.183	0.496	0.240	0.439
	scaffold	43	0.207	0.547	0.305	0.532
	scaffold	44	0.185	0.528	0.267	0.521
	temporal	42	0.245	0.608	0.403	0.572
	temporal	43	0.225	0.573	0.347	0.514
	temporal	44	0.228	0.566	0.350	0.515
MLP	cold_condition	42	0.271	0.577	0.369	0.520
	cold_condition	43	0.270	0.582	0.374	0.528
	cold_condition	44	0.266	0.576	0.369	0.520
	cold_drug	42	0.276	0.546	0.339	0.496
	cold_drug	43	0.287	0.551	0.344	0.511
	cold_drug	44	0.243	0.495	0.292	0.435
	degree_balanced	42	0.362	0.630	0.445	0.586
	degree_balanced	43	0.354	0.622	0.435	0.579
	degree_balanced	44	0.347	0.615	0.423	0.568
	random	42	0.347	0.611	0.418	0.569
	random	43	0.368	0.630	0.447	0.594
	random	44	0.358	0.617	0.430	0.573
	scaffold	42	0.204	0.531	0.266	0.517
	scaffold	43	0.186	0.498	0.226	0.482
	scaffold	44	0.198	0.530	0.270	0.542
	temporal	42	0.202	0.517	0.284	0.454
	temporal	43	0.210	0.553	0.311	0.498
	temporal	44	0.212	0.544	0.304	0.486
XGBoost	cold_condition	42–44	0.338	0.686	0.570	0.725
	cold_drug	42–44	0.414	0.683	0.555	0.715
	degree_balanced	42–44	0.521	0.758	0.645	0.776
	random	42–44	0.510	0.751	0.637	0.771
	scaffold	42–44	0.193	0.567	0.374	0.640
	temporal	42–44	0.193	0.602	0.454	0.669

B.4 PPI ML Results (54 runs, aggregated)

C Complete LLM Results

This appendix presents complete LLM results for all 241 experiments. DTI uses 6 models (81 runs); CT and PPI each use 5 models (80 runs each). Each model is evaluated in zero-shot and three 3-shot configurations (different random example sets). 3-shot results report mean $\pm$ std across the three example sets.

C.1 DTI LLM Results (81 runs)

DTI models: Gemini-2.5-Flash, Gemini-2.5-Flash-Lite, GPT-4o-mini, Llama-3.3-70B, Mistral-7B, Qwen2.5-32B. L2 was not evaluated for DTI due to annotation cost.

Table 7: PPI ML results: 3 models × 6 configurations, mean ± std over 3 seeds.

Model	Split	Negatives	LogAUC	AUPRC	MCC	AUROC
SiameseCNN	random	negbiodb	.517±.018	.961±.001	.794±.012	.963±.000
SiameseCNN	random	unif_rand	.552±.002	.964±.001	.806±.007	.965±.001
SiameseCNN	random	deg_match	.548±.011	.963±.001	.803±.005	.964±.001
SiameseCNN	cold_protein	negbiodb	.314±.014	.880±.003	.568±.019	.873±.002
SiameseCNN	cold_both	negbiodb	.037±.010	.702±.031	.070±.004	.585±.040
SiameseCNN	ddb	negbiodb	.534±.011	.961±.001	.795±.004	.962±.001
PIPR	random	negbiodb	.519±.009	.962±.000	.812±.006	.964±.001
PIPR	random	unif_rand	.565±.005	.966±.000	.810±.002	.966±.000
PIPR	random	deg_match	.550±.009	.965±.001	.817±.006	.966±.001
PIPR	cold_protein	negbiodb	.288±.010	.869±.006	.565±.019	.859±.008
PIPR	cold_both	negbiodb	.031±.019	.610±.055	−.018±.044	.409±.077
PIPR	ddb	negbiodb	.537±.009	.962±.000	.808±.003	.964±.000
MLPFeatures	random	negbiodb	.567±.005	.962±.001	.788±.003	.962±.001
MLPFeatures	random	unif_rand	.539±.175	.949±.043	.766±.118	.948±.044
MLPFeatures	random	deg_match	.458±.059	.934±.013	.716±.033	.930±.012
MLPFeatures	cold_protein	negbiodb	.476±.005	.935±.001	.706±.005	.931±.001
MLPFeatures	cold_both	negbiodb	.595±.051	.973±.010	.749±.043	.950±.021
MLPFeatures	ddb	negbiodb	.564±.005	.961±.000	.787±.001	.961±.000

Table 8: DTI L1 (MCQ, 4-way classification). N=3 for 3-shot (mean±std), N=1 for zero-shot.

Model	Config	Accuracy	Macro-F1	MCC
Gemini-2.5-Flash	3-shot	1.000±0.000	1.000±0.000	1.000±0.000
Gemini-2.5-Flash	zero-shot	0.721	0.630	0.610
Gemini-2.5-Flash-Lite	3-shot	0.971±0.019	0.964±0.031	0.960±0.026
Gemini-2.5-Flash-Lite	zero-shot	0.807	0.705	0.750
GPT-4o-mini	3-shot	0.944±0.012	0.941±0.014	0.922±0.016
GPT-4o-mini	zero-shot	0.736	0.642	0.632
Llama-3.3-70B	3-shot	0.991±0.010	0.991±0.010	0.987±0.014
Llama-3.3-70B	zero-shot	0.613	0.501	0.430
Mistral-7B	3-shot	0.650±0.060	0.496±0.108	0.502±0.086
Mistral-7B	zero-shot	0.708	0.551	0.593
Qwen2.5-32B	3-shot	0.977±0.012	0.978±0.012	0.968±0.017
Qwen2.5-32B	zero-shot	0.728	0.635	0.620

Table 9: DTI L3 (reasoning, LLM-as-judge). Judge dimensions: accuracy, completeness, reasoning quality, specificity. Overall = mean of 4 dimensions.

Model	Config	Accuracy	Complete.	Reasoning	Specific.	Overall
Gemini-2.5-Flash	3-shot	4.88±0.01	4.49±0.08	4.89±0.02	4.22±0.01	4.62±0.02
Gemini-2.5-Flash	zero-shot	4.85	4.70	4.80	4.30	4.66
Gemini-Lite	3-shot	4.31±0.03	4.06±0.03	4.28±0.03	3.23±0.05	3.97±0.02
Gemini-Lite	zero-shot	4.20	4.05	4.10	3.08	3.86
GPT-4o-mini	3-shot	4.06±0.01	4.04±0.02	4.02±0.02	2.59±0.04	3.68±0.02
GPT-4o-mini	zero-shot	3.97	3.97	3.97	2.85	3.69
Llama-3.3-70B	3-shot	4.18±0.06	4.10±0.04	4.03±0.04	2.41±0.07	3.68±0.05
Llama-3.3-70B	zero-shot	3.94	3.86	3.72	2.36	3.47
Mistral-7B	3-shot	3.46±0.21	3.63±0.15	3.23±0.05	1.77±0.10	3.02±0.08
Mistral-7B	zero-shot	3.55	3.66	3.48	2.24	3.23
Qwen2.5-32B	3-shot	4.04±0.08	4.02±0.06	3.90±0.07	2.66±0.01	3.65±0.05
Qwen2.5-32B	zero-shot	3.97	3.97	3.87	2.82	3.66

Table 10: DTI L4 (discrimination, tested vs. untested). Evidence citation rate measures hallucination.

Model	Config	Accuracy	MCC	Contam. Gap	Halluc. Rate
Gemini-2.5-Flash	3-shot	0.478±0.006	−0.102±0.005	0.033±0.036	1.000
Gemini-2.5-Flash	zero-shot	0.427	−0.234	0.047	0.994
Gemini-Lite	3-shot	0.570±0.040	0.181±0.087	0.022±0.027	0.990±0.008
Gemini-Lite	zero-shot	0.350	−0.349	0.146	0.727
GPT-4o-mini	3-shot	0.512±0.011	0.037±0.036	0.249±0.011	0.991±0.012
GPT-4o-mini	zero-shot	0.517	0.047	0.233	1.000
Llama-3.3-70B	3-shot	0.589±0.026	0.184±0.051	0.242±0.014	1.000
Llama-3.3-70B	zero-shot	0.540	0.101	−0.024	1.000
Mistral-7B	3-shot	0.491±0.013	−0.030±0.042	0.049±0.047	1.000
Mistral-7B	zero-shot	0.500	0.000	0.000	1.000
Qwen2.5-32B	3-shot	0.538±0.013	0.113±0.038	0.163±0.021	1.000
Qwen2.5-32B	zero-shot	0.510	0.046	0.098	1.000

C.2 CT LLM Results (80 runs)

CT models: Claude Haiku-4.5, Gemini-2.5-Flash, GPT-4o-mini, Llama-3.3-70B, Qwen2.5-32B.

Table 11: CT-L1 (MCQ, 5-way failure classification).

Model	Config	Accuracy	Macro-F1	MCC
Claude Haiku-4.5	3-shot	0.662±0.012	0.657±0.015	0.592±0.014
Claude Haiku-4.5	zero-shot	0.660	0.652	0.581
Gemini-2.5-Flash	3-shot	0.667±0.014	0.663±0.017	0.597±0.015
Gemini-2.5-Flash	zero-shot	0.681	0.675	0.609
GPT-4o-mini	3-shot	0.625±0.011	0.616±0.012	0.546±0.012
GPT-4o-mini	zero-shot	0.641	0.634	0.571
Llama-3.3-70B	3-shot	0.634±0.022	0.630±0.030	0.560±0.026
Llama-3.3-70B	zero-shot	0.631	0.617	0.559
Qwen2.5-32B	3-shot	0.648±0.017	0.642±0.022	0.572±0.024
Qwen2.5-32B	zero-shot	0.654	0.641	0.579

Table 12: CT-L2 (structured extraction from clinical trial evidence).

Model	Config	Category Acc	Field F1	Schema Compl.
Claude Haiku-4.5	3-shot	0.738±0.055	0.476±0.099	1.000
Claude Haiku-4.5	zero-shot	0.725	0.280	1.000
Gemini-2.5-Flash	3-shot	0.742±0.068	0.746±0.162	1.000
Gemini-2.5-Flash	zero-shot	0.760	0.284	1.000
GPT-4o-mini	3-shot	0.715±0.089	0.734±0.185	0.917±0.001
GPT-4o-mini	zero-shot	0.751	0.334	0.965
Llama-3.3-70B	3-shot	0.752±0.064	0.768±0.161	1.000
Llama-3.3-70B	zero-shot	0.762	0.315	1.000
Qwen2.5-32B	3-shot	0.709±0.095	0.808±0.162	1.000
Qwen2.5-32B	zero-shot	0.718	0.315	1.000

Table 13: CT-L3 (reasoning, LLM-as-judge overall score, 1–5 scale).

Model	Config	Overall Score
Claude Haiku-4.5	3-shot	4.960±0.007
Claude Haiku-4.5	zero-shot	5.000
Gemini-2.5-Flash	3-shot	4.462±0.050
Gemini-2.5-Flash	zero-shot	5.000
GPT-4o-mini	3-shot	4.743±0.058
GPT-4o-mini	zero-shot	4.661
Llama-3.3-70B	3-shot	4.826±0.007
Llama-3.3-70B	zero-shot	4.997
Qwen2.5-32B	3-shot	4.968±0.007
Qwen2.5-32B	zero-shot	5.000

Table 14: CT-L4 (discrimination, tested vs. untested clinical trials).

Model	Config	Accuracy	MCC	Halluc. Rate
Claude Haiku-4.5	3-shot	0.739±0.019	0.502±0.014	1.000
Claude Haiku-4.5	zero-shot	0.750	0.514	1.000
Gemini-2.5-Flash	3-shot	0.777±0.011	0.563±0.018	1.000
Gemini-2.5-Flash	zero-shot	0.748	0.496	1.000
GPT-4o-mini	3-shot	0.738±0.008	0.485±0.007	1.000
GPT-4o-mini	zero-shot	0.744	0.491	1.000
Llama-3.3-70B	3-shot	0.739±0.023	0.504±0.036	1.000
Llama-3.3-70B	zero-shot	0.635	0.364	1.000
Qwen2.5-32B	3-shot	0.724±0.017	0.484±0.018	1.000
Qwen2.5-32B	zero-shot	0.757	0.519	1.000

C.3 PPI LLM Results (80 runs)

PPI models: Claude Haiku-4.5, Gemini-2.5-Flash, GPT-4o-mini, Llama-3.3-70B, Qwen2.5-32B.

Table 15: PPI-L1 (MCQ, 4-way evidence quality classification). All 3-shot models achieve $\geq$0.999 accuracy except Qwen2.5-32B. Zero-shot performance drops to 0.75 due to complete failure on direct_experimental evidence (Class A: 0.0 accuracy), while scoring $\approx$1.0 on Classes B–D.

Model	Config	Accuracy	Macro-F1	MCC
Claude Haiku-4.5	3-shot	0.999±0.001	0.999±0.001	0.999±0.001
Claude Haiku-4.5	zero-shot	0.750	0.667	0.730
Gemini-2.5-Flash	3-shot	1.000±0.001	1.000±0.001	0.999±0.001
Gemini-2.5-Flash	zero-shot	0.750	0.667	0.730
GPT-4o-mini	3-shot	1.000±0.001	1.000±0.001	0.999±0.001
GPT-4o-mini	zero-shot	0.749	0.665	0.728
Llama-3.3-70B	3-shot	1.000	1.000	1.000
Llama-3.3-70B	zero-shot	0.750	0.667	0.730
Qwen2.5-32B	3-shot	0.826±0.069	0.792±0.101	0.803±0.069
Qwen2.5-32B	zero-shot	0.750	0.667	0.730

Table 16: PPI-L2 (extraction, protein pair identification). Near-perfect entity and count extraction; method and interaction strength require explicit evidence (zero-shot method accuracy is 0.000 for all models).

Model	Config	Entity F1	Count Acc	Schema Compl.	Method Acc	Strength Acc
Claude Haiku-4.5	3-shot	1.000	1.000	1.000	0.088±0.139	0.522±0.151
Claude Haiku-4.5	zero-shot	1.000	1.000	1.000	0.000	0.594
Gemini-2.5-Flash	3-shot	1.000	1.000	1.000	1.000	0.399±0.219
Gemini-2.5-Flash	zero-shot	0.952	1.000	0.902	0.000	0.554
GPT-4o-mini	3-shot	0.999±0.001	1.000	1.000	0.937±0.108	0.425±0.182
GPT-4o-mini	zero-shot	0.999	1.000	1.000	0.000	0.307
Llama-3.3-70B	3-shot	1.000	1.000	1.000	0.082±0.143	0.608±0.019
Llama-3.3-70B	zero-shot	1.000	1.000	1.000	0.000	0.572
Qwen2.5-32B	3-shot	0.998±0.001	1.000	1.000	0.938±0.108	0.501±0.166
Qwen2.5-32B	zero-shot	0.999	1.000	1.000	0.000	0.510

Table 17: PPI-L3 (reasoning, LLM-as-judge overall score, 1–5 scale). Llama-3.3-70B 3-shot has 51.5% error rate; score shown for successful completions only (†).

Model	Config	Overall Score
Claude Haiku-4.5	3-shot	3.70±0.25
Claude Haiku-4.5	zero-shot	4.683
Gemini-2.5-Flash	3-shot	3.11±0.10
Gemini-2.5-Flash	zero-shot	4.645
GPT-4o-mini	3-shot	3.21±0.10
GPT-4o-mini	zero-shot	4.361
Llama-3.3-70B	3-shot	2.05±0.92†
Llama-3.3-70B	zero-shot	4.281
Qwen2.5-32B	3-shot	3.51±0.04
Qwen2.5-32B	zero-shot	4.452

D L3 Reasoning Analysis

L3 evaluates open-ended scientific reasoning about negative results using an LLM-as-judge (GPT-4o-mini) scoring on four dimensions (1–5 scale). We present the full per-dimension breakdown and discuss the ceiling effect that motivated deferring L3 from the main text.

D.1 Judge Ceiling Effect

The GPT-4o-mini judge exhibits a severe ceiling effect in two domains:

- **CT**: Overall scores range 4.46–5.00 (Table 13). Four of five models receive perfect 5.0 scores in zero-shot mode. This ceiling confounds meaningful cross-model comparison.

Table 18: PPI-L4 (discrimination, tested vs. untested protein pairs). Cit. Rate = fraction of responses including any evidence citation (all cited evidence is fabricated; hallucination rate = 100%).

Model	Config	Accuracy	MCC	Cit. Rate
Claude Haiku-4.5	3-shot	0.648±0.010	0.390±0.020	1.000
Claude Haiku-4.5	zero-shot	0.608	0.334	1.000
Gemini-2.5-Flash	3-shot	0.671±0.006	0.382±0.004	1.000
Gemini-2.5-Flash	zero-shot	0.647	0.358	1.000
GPT-4o-mini	3-shot	0.633±0.025	0.352±0.039	0.888±0.036
GPT-4o-mini	zero-shot	0.699	0.430	1.000
Llama-3.3-70B	3-shot	0.637±0.046	0.371±0.056	0.978±0.001
Llama-3.3-70B	zero-shot	0.703	0.441	1.000
Qwen2.5-32B	3-shot	0.641±0.010	0.369±0.009	0.467±0.032
Qwen2.5-32B	zero-shot	0.645	0.366	1.000

- **PPI**: Zero-shot scores range 4.28–4.68 (Table 17). While lower than CT, the range is still compressed. Notably, Llama-3.3-70B 3-shot has a 51.5% formatting error rate; successful completions score 2.05±0.92, but overall 3-shot performance is unreliable.

DTI L3 shows the most meaningful variation (3.02–4.66), with Gemini-2.5-Flash achieving the highest overall score.

D.2 PPI L3 Per-Dimension Scores

Table 19 reveals that **structural reasoning** (dimension 2) is the most challenging aspect across all models. Even top-performing models score 1.2–4.4 on structural reasoning while achieving 4.6–5.0 on biological plausibility. This suggests models excel at identifying functional mismatches between proteins but struggle with detailed structural arguments about binding interfaces, domain compatibility, and steric constraints.

Table 19: PPI L3 per-dimension judge scores. Structural reasoning is consistently the weakest dimension. †Llama-3.3-70B 3-shot has 51.5% output failure rate; scores computed from successful completions only.

Model	Config	Bio. Plaus.	Struct. Reas.	Mech. Compl.	Specificity	Overall
Claude Haiku-4.5	3-shot	4.98	2.25	3.17	4.39	3.70
Claude Haiku-4.5	zero-shot	5.00	4.29	4.46	4.99	4.68
Gemini-2.5-Flash	3-shot	4.71	1.24	2.70	3.77	3.11
Gemini-2.5-Flash	zero-shot	5.00	4.39	4.19	5.00	4.65
GPT-4o-mini	3-shot	4.80	1.34	2.78	3.92	3.21
GPT-4o-mini	zero-shot	4.87	3.98	3.87	4.73	4.36
Llama-3.3-70B	3-shot	2.81	1.18	1.80	2.43	2.05†
Llama-3.3-70B	zero-shot	4.77	3.94	3.70	4.71	4.28
Qwen2.5-32B	3-shot	4.92	1.96	3.06	4.10	3.51
Qwen2.5-32B	zero-shot	4.91	4.01	4.02	4.87	4.45

D.3 3-shot Degradation in PPI L3

A notable phenomenon: 3-shot prompting *degrades* PPI L3 scores compared to zero-shot for all models (e.g., Gemini: 4.65→3.11, GPT-4o-mini: 4.36→3.21). This contrasts with L1 where 3-shot dramatically improves performance. We hypothesize that few-shot examples constrain the model's reasoning to follow a specific template, reducing the depth and specificity of explanations compared to unconstrained zero-shot generation. This effect is most extreme for Llama-3.3-70B, where 3-shot examples cause a 51.5% output failure rate—successful completions score only 2.05±0.92.

D.4 Cross-Domain L3 Comparison

Despite the ceiling effect, two patterns emerge:

1. **Gemini-2.5-Flash consistently leads** across DTI (4.62 3-shot) and PPI (4.65 zero-shot), producing the most scientifically specific explanations.
2. **Specificity is the hardest dimension in DTI** (2.41–4.22) but not in PPI (3.77–5.00), suggesting that DTI requires more specialized pharmacological knowledge that models lack.

E Full PPI Contamination Analysis

This appendix presents the complete PPI L4 contamination analysis, including temporal stratification and protein popularity controls.

E.1 Temporal Contamination

Table 20 shows per-run accuracy for pre-2015 and post-2020 IntAct interaction pairs across all 20 PPI-L4 runs (5 models $\times$ 4 configs). All 20 runs exceed the 0.15 contamination threshold [Balloccu et al., 2024], with gaps ranging from 0.361 to 0.612.

Table 20: PPI L4 temporal contamination: accuracy on pre-2015 vs. post-2020 pairs. Gap > 0.15 indicates likely memorization. All runs flagged.

Model	Config	Acc pre-2015	Acc post-2020	Gap	Flag
Claude Haiku-4.5	3-shot (set 0)	0.598	0.041	0.557	YES
Claude Haiku-4.5	3-shot (set 1)	0.618	0.051	0.567	YES
Claude Haiku-4.5	3-shot (set 2)	0.569	0.031	0.538	YES
Claude Haiku-4.5	zero-shot	0.422	0.020	0.401	YES
Gemini-2.5-Flash	3-shot (set 0)	0.765	0.184	0.581	YES
Gemini-2.5-Flash	3-shot (set 1)	0.686	0.133	0.554	YES
Gemini-2.5-Flash	3-shot (set 2)	0.706	0.184	0.522	YES
Gemini-2.5-Flash	zero-shot	0.588	0.133	0.456	YES
GPT-4o-mini	3-shot (set 0)	0.569	0.112	0.456	YES
GPT-4o-mini	3-shot (set 1)	0.422	0.051	0.371	YES
GPT-4o-mini	3-shot (set 2)	0.569	0.092	0.477	YES
GPT-4o-mini	zero-shot	0.762	0.245	0.517	YES
Llama-3.3-70B	3-shot (set 0)	0.422	0.041	0.381	YES
Llama-3.3-70B	3-shot (set 1)	0.745	0.133	0.612	YES
Llama-3.3-70B	3-shot (set 2)	0.402	0.041	0.361	YES
Llama-3.3-70B	zero-shot	0.794	0.204	0.590	YES
Qwen2.5-32B	3-shot (set 0)	0.588	0.112	0.476	YES
Qwen2.5-32B	3-shot (set 1)	0.510	0.061	0.449	YES
Qwen2.5-32B	3-shot (set 2)	0.529	0.082	0.448	YES
Qwen2.5-32B	zero-shot	0.598	0.071	0.527	YES

E.2 Contamination vs. Protein Popularity

A potential confound: pre-2015 proteins may be better-studied (higher degree in interaction networks), and models might simply know more about popular proteins regardless of memorization. To control for this, we stratify L4 accuracy by protein pair degree (median split at degree 172.2).

Table 21 shows that the temporal gap persists in *both* high-degree and low-degree protein pairs for all five models. Model-averaged gaps:

- **High-degree pairs**: gaps 0.44–0.62 (all $\gg 0.15$)
- **Low-degree pairs**: gaps 0.30–0.57 (all $\gg 0.15$)

This confirms that the temporal signal reflects genuine memorization of interaction databases, not a popularity confound. Notably, Gemini-2.5-Flash shows a *stronger* gap for low-degree pairs (0.53 vs. 0.50), suggesting it has particularly memorized obscure protein interactions.

E.3 Model-Averaged Summary

Key finding: All five models show contamination gaps exceeding the 0.15 threshold in both high-degree and low-degree strata. The contamination signal is robust to protein popularity and reflects genuine memorization of interaction databases rather than a confound with protein familiarity.

F LLM Prompt Examples

This appendix shows representative prompt templates for each evaluation level across all three domains. Each domain uses a domain-specific system prompt and task-specific user prompts. All prompts follow a zero-shot and 3-shot structure.

Table 21: PPI L4 contamination stratified by protein popularity. Gap persists for both high-degree and low-degree pairs, confirming true memorization.

Model	Config	Pre-2015 Acc		Post-2020 Acc		Gap	
		High	Low	High	Low	High	Low
Haiku-4.5	3-shot	0.661	0.514	0.045	0.037	0.615	0.478
Haiku-4.5	zero-shot	0.518	0.304	0.045	0.000	0.472	0.304
Gemini	3-shot	0.768	0.660	0.250	0.099	0.518	0.561
Gemini	zero-shot	0.643	0.522	0.205	0.074	0.438	0.448
GPT-4o-mini	3-shot	0.637	0.377	0.099	0.074	0.539	0.303
GPT-4o-mini	zero-shot	0.855	0.652	0.250	0.241	0.605	0.411
Llama-3.3	3-shot	0.643	0.377	0.129	0.025	0.514	0.352
Llama-3.3	zero-shot	0.857	0.717	0.273	0.148	0.584	0.569
Qwen2.5	3-shot	0.655	0.406	0.129	0.050	0.526	0.357
Qwen2.5	zero-shot	0.679	0.500	0.091	0.056	0.588	0.444

Table 22: Model-averaged contamination gaps by protein popularity.

Model	Avg Gap (High)	Avg Gap (Low)	Verdict
Claude Haiku-4.5	0.580	0.434	True contamination
Gemini-2.5-Flash	0.498	0.532	True contamination (stronger for obscure)
GPT-4o-mini	0.555	0.330	True contamination
Llama-3.3-70B	0.532	0.406	True contamination
Qwen2.5-32B	0.541	0.378	True contamination

F.1 System Prompts

```
DTI: "You are a pharmaceutical scientist with expertise in drug-target
interactions, assay development, and medicinal chemistry. Provide precise,
evidence-based answers."

CT:  "You are a clinical trial expert with deep knowledge of drug
development, regulatory science, and clinical pharmacology."

PPI: "You are a protein biochemist with expertise in protein-protein
interactions, structural biology, and proteomics experimental methods.
Provide precise, evidence-based answers."
```

F.2 L1: Multiple Choice Classification

DTI L1 (4-way: hard negative / conditional negative / methodological negative / dose-time negative):

```
[context_text with evidence description]

Respond with ONLY the letter of the correct answer: A, B, C, or D.
```

CT L1 (5-way: safety / efficacy / enrollment / strategic / other):

```
Based on the clinical trial information below, classify the primary
reason for this trial's failure.

[context_text with trial details]

Categories:
A) Safety -- Trial failed due to drug toxicity, adverse events, or
   safety signals
B) Efficacy -- Trial failed to demonstrate therapeutic benefit vs control
C) Enrollment -- Trial failed to recruit sufficient participants
D) Strategic -- Trial was discontinued for business, strategic, or
   portfolio reasons
E) Other -- Trial failed due to study design flaws, regulatory issues,
   or other reasons

Respond with ONLY a single letter (A, B, C, D, or E).
```

PPI L1 (4-way: direct experimental / systematic screen / computational inference / database score absence):

```
Based on the evidence description below, classify the type and quality
of evidence supporting this protein non-interaction.

[context_text with evidence description]

Categories:
A) Direct experimental -- A specific binding assay (co-IP, pulldown,
   SPR, etc.) found no physical interaction
B) Systematic screen -- A high-throughput binary screen (Y2H, LUMIER,
   etc.) found no interaction
C) Computational inference -- ML analysis of co-fractionation or complex
   data predicts no interaction
D) Database score absence -- Zero or negligible combined interaction
   score across multiple evidence channels

Respond with ONLY a single letter (A, B, C, or D).
```

F.3 L2: Structured Extraction

DTI L2 (JSON extraction from abstract):

```
Extract all negative drug-target interaction results from the following
abstract.

Abstract: [abstract_text]

For each negative result found, extract:
- compound: compound/drug name
- target: target protein/gene name
- target_uniprot: UniProt accession (if determinable)
- activity_type: type of measurement (IC50, Ki, Kd, EC50, etc.)
- activity_value: reported value with units
- activity_relation: relation (=, >, <, ~)
- assay_format: biochemical, cell-based, or in vivo
- outcome: inactive, weak, or inconclusive

Also report:
- total_inactive_count: total number of inactive results mentioned
- positive_results_mentioned: true/false

Respond in JSON format.
```

CT L2 (JSON extraction from termination report):

```
Extract structured failure information from the following clinical trial
termination report. Return a JSON object with the fields specified below.

[context_text]

Required JSON fields:
- failure_category: one of [efficacy, safety, pharmacokinetic,
  enrollment, strategic, design, regulatory, other]
- failure_subcategory: specific reason
- affected_system: organ system affected (null if not applicable)
- severity_indicator: one of [mild, moderate, severe, fatal, null]
- quantitative_evidence: true if text mentions specific numbers
- decision_maker: who terminated [sponsor, dsmb, regulatory,
  investigator, null]
- patient_impact: brief description of patient safety impact

Return ONLY valid JSON, no additional text.
```

PPI L2 (JSON extraction of non-interacting pairs):

```
Extract all protein pairs reported as non-interacting from the following
evidence summary. Return a JSON object with the fields specified below.

[context_text]

Required JSON fields:
- non_interacting_pairs: list of objects, each with:
    - protein_1: gene symbol or UniProt accession
    - protein_2: gene symbol or UniProt accession
    - method: experimental method used
    - evidence_strength: one of [strong, moderate, weak]
- total_negative_count: total number of non-interacting pairs mentioned
- positive_interactions_mentioned: true if any positive interactions
  are also mentioned

Return ONLY valid JSON, no additional text.
```

F.4 L3: Scientific Reasoning

DTI L3:

```
[context_text describing compound-target pair and inactivity evidence]

Provide a detailed scientific explanation (3-5 paragraphs) covering:
1. Structural compatibility between compound and target binding site
2. Known selectivity profile and mechanism of action
3. Relevant SAR (structure-activity relationship) data
4. Pharmacological context and therapeutic implications
```

CT L3:

```
The following clinical trial was confirmed as a FAILURE. Based on the
trial data below, provide a scientific explanation for why this drug
failed in this clinical trial.

[context_text]

Your explanation should address:
1. Mechanism -- What is the drug's mechanism of action and why might
   it be insufficient for this condition?
2. Evidence interpretation -- What do the statistical results tell us?
3. Clinical factors -- Trial design, patient population, or disease
   biology factors?
4. Broader context -- Known challenges of treating this condition?

Provide a thorough explanation in 3-5 paragraphs.
```

PPI L3:

```
The following two proteins have been experimentally tested and confirmed
to NOT physically interact. Based on the protein information below,
provide a scientific explanation for why they are unlikely to form a
physical interaction.

[context_text]

Your explanation should address:
1. Biological plausibility -- Are there biological reasons (function,
   pathway, localization) that make interaction unlikely?
2. Structural reasoning -- Do domain architectures, binding interfaces,
   or steric factors argue against interaction?
3. Mechanistic completeness -- Are multiple relevant factors considered?
4. Specificity -- Are claims specific to these proteins or generic?

Provide a thorough explanation in 3-5 paragraphs.
```

F.5 L4: Tested vs. Untested Discrimination

All three domains use a similar L4 structure:

```
DTI: "Has the following compound-target pair been experimentally
tested for binding activity?"
[compound name + target name]

CT:  "Has the following drug-condition combination ever been tested
in a registered clinical trial?"
[drug name + condition name]

PPI: "Has the following protein pair ever been experimentally tested
for physical interaction?"
[protein A gene symbol + protein B gene symbol]

Response format (all domains):
"On the first line, respond with ONLY 'tested' or 'untested'.
On the second line, provide brief evidence for your answer."
```

F.6 Few-Shot Configuration

For 3-shot prompting, three independent example sets are sampled (seeds 42, 43, 44), and results are reported as mean $\pm$ std across the three sets. Examples are formatted as:

```
Here are examples of [task description]:

[Example 1 context]
Answer: [gold answer]

---

[Example 2 context]
Answer: [gold answer]

---

[Example 3 context]
Answer: [gold answer]

Now [task instruction]:

[Test instance context]
```

G Database Schema

NegBioDB uses three separate SQLite databases, one per domain, sharing common design patterns: WAL journal mode, foreign key enforcement, COALESCE-based deduplication indexes, and a four-tier confidence system (gold/silver/bronze/copper). Full DDL for all migrations is available in the repository. Below we summarize the key tables.

G.1 DTI Domain Schema

Two migrations: `001_initial_schema` (core tables) and `002_target_variants` (variant support).

```
-- Core entity tables
compounds       (compound_id PK, canonical_smiles, inchikey UNIQUE,
                 inchikey_connectivity, pubchem_cid, chembl_id,
                 molecular_weight, logp, hbd, hba, tpsa, qed, ...)

targets         (target_id PK, uniprot_accession UNIQUE,
                 chembl_target_id, gene_symbol, target_family,
```

```
                  development_level CHECK IN (Tclin/Tchem/Tbio/Tdark), ...)

assays           (assay_id PK, source_db, source_assay_id,
                  assay_format CHECK IN (biochemical/cell-based/in_vivo),
                  screen_type, z_factor, pubmed_id, ...)

-- Core fact table (30.5M rows)
negative_results (result_id PK, compound_id FK, target_id FK, assay_id FK,
                   result_type CHECK IN (hard_negative/conditional_negative/
                     methodological_negative/dose_time_negative/
                     hypothesis_negative),
                   confidence_tier CHECK IN (gold/silver/bronze/copper),
                   activity_type, activity_value, pchembl_value,
                   source_db, source_record_id, extraction_method, ...)

-- Dedup: UNIQUE(compound_id, target_id, COALESCE(assay_id,-1),
--               source_db, source_record_id)

-- Aggregation (for ML export)
compound_target_pairs (pair_id PK, compound_id FK, target_id FK,
                       num_assays, num_sources, best_confidence,
                       compound_degree, target_degree, ...)

-- Variant support (migration 002)
target_variants (variant_id PK, target_id FK, variant_label,
                 source_db, UNIQUE(target_id, variant_label, ...))
```

G.2 CT Domain Schema

Two migrations: `001_ct_initial_schema` (core tables) and `002_schema_fixes` (expert review fixes).

```
-- Entity tables
interventions   (intervention_id PK, intervention_type CHECK IN
                  (drug/biologic/device/...),
                 intervention_name, chembl_id, canonical_smiles,
                 inchikey, molecular_type, ...)

conditions      (condition_id PK, condition_name, mesh_id,
                 icd10_code, therapeutic_area, ...)

clinical_trials (trial_id PK, source_trial_id UNIQUE,
                 overall_status, trial_phase, enrollment_actual,
                 primary_endpoint, why_stopped,
                 termination_type CHECK IN (clinical_failure/
                   administrative/external_event/unknown), ...)

-- Core fact table (132,925 rows)
trial_failure_results (result_id PK, intervention_id FK,
                       condition_id FK, trial_id FK,
  failure_category CHECK IN (efficacy/safety/pharmacokinetic/
    enrollment/strategic/regulatory/design/other),
  confidence_tier CHECK IN (gold/silver/bronze/copper),
  p_value_primary, effect_size, serious_adverse_events,
  highest_phase_reached, result_interpretation CHECK IN
    (definitive_negative/inconclusive_underpowered/
     mixed_endpoints/futility_stopped/safety_stopped/
     administrative),
  source_db, extraction_method, ...)

-- Dedup: UNIQUE(intervention_id, condition_id,
--               COALESCE(trial_id,-1), source_db, source_record_id)

-- Junction tables
```

```
trial_interventions (trial_id FK, intervention_id FK, arm_role)
trial_conditions    (trial_id FK, condition_id FK)
intervention_targets (intervention_id FK, uniprot_accession, ...)
```

G.3 PPI Domain Schema

Two migrations: `001_ppi_initial_schema` (core tables) and `002_llm_annotations` (protein annotations for LLM benchmark).

```
-- Entity table
proteins        (protein_id PK, uniprot_accession UNIQUE,
                 gene_symbol, amino_acid_sequence, sequence_length,
                 subcellular_location,
                 function_description, go_terms,
                 domain_annotations, ...)  -- migration 002

-- Core fact table (2.23M rows)
ppi_negative_results (result_id PK, protein1_id FK, protein2_id FK,
  experiment_id FK,
  evidence_type CHECK IN (experimental_non_interaction/
    ml_predicted_negative/low_score_negative/
    compartment_separated/literature_reported),
  confidence_tier CHECK IN (gold/silver/bronze/copper),
  interaction_score, detection_method,
  source_db, extraction_method, ...,
  CHECK (protein1_id < protein2_id))  -- canonical ordering

-- Dedup: UNIQUE(protein1_id, protein2_id,
--               COALESCE(experiment_id,-1),
--               source_db, source_record_id)

-- Aggregation
protein_protein_pairs (pair_id PK, protein1_id FK, protein2_id FK,
                       num_experiments, num_sources, best_confidence,
                       protein1_degree, protein2_degree, ...,
                       CHECK (protein1_id < protein2_id))

-- LLM support (migration 002)
ppi_publication_abstracts (pmid PK, title, abstract, ...)
```

G.4 Common Design Patterns

- **Deduplication:** All fact tables use `COALESCE(fk, -1)` in UNIQUE indexes to handle NULL foreign keys (SQLite treats NULLs as distinct in UNIQUE constraints).
- **Confidence tiers:** Four-level system across all domains: gold (systematic screens, multiple confirmations) > silver (ML-derived, p-value based) > bronze (computational, NLP-detected) > copper (label-only).
- **Aggregation tables:** Pre-computed pair-level statistics for ML export, avoiding expensive JOINs during dataset construction.
- **Symmetric pairs (PPI):** `CHECK (protein1_id < protein2_id)` enforces canonical ordering, preventing duplicate pair representations.
- **Schema migrations:** All databases track applied migrations in a `schema_migrations` table for reproducible upgrades.

H Splitting Strategy Details

This appendix describes the splitting strategies used across the three domains and their implementation details.

Table 23: Split strategies by domain. ✓= implemented, — = not applicable.

Strategy	DTI	CT	PPI	Description
Random	✓	✓	✓	Stratified random assignment (70/10/20)
Cold_compound/drug	✓	✓	—	All pairs with held-out compounds in test
Cold_target/condition	✓	✓	—	All pairs with held-out targets in test
Cold_protein	—	—	✓	All pairs with held-out proteins in test
Cold_both	—	—	✓	METIS graph partitioning; unseen proteins on both sides
Temporal	—	✓	—	$\leq$2017 train, 2018–19 val, $\geq$2020 test
Scaffold	—	✓	—	Murcko scaffold-based grouping
DDB	✓	—	✓	Degree-balanced binning

H.1 Split Strategy Overview

H.2 Cold Splitting

Cold compound/drug/protein. Entities are randomly partitioned into train/val/test groups. All pairs containing a held-out entity are assigned to the corresponding fold. This tests generalization to unseen chemical or biological entities.

Cold_both (PPI only). We use METIS graph partitioning [Karypis and Kumar, 1998] to partition proteins into three groups such that proteins in the test set have no interactions with proteins in the training set. This creates a maximally challenging generalization test where *both* proteins in a test pair are unseen during training. Implementation uses the `pymetis` library with k=3 partitions, targeting 70/10/20 splits. The resulting test partition contains only 1.7% positive examples (242/14,037) due to the extreme network separation, creating a highly imbalanced evaluation setting.

H.3 Temporal Splitting (CT only)

Clinical trials are split by primary completion date: trials completing $\leq$2017 form the training set (42,676 pairs), 2018–2019 form validation (9,257 pairs), and $\geq$2020 form the test set (50,917 pairs). This mimics a realistic prospective prediction scenario. A known limitation: the temporal split can produce single-class validation sets (all negative) for CT-M1, since successful trials are rare in certain time windows. When this occurs, AUROC and other threshold-dependent metrics are undefined.

H.4 Scaffold Splitting (CT only)

For interventions with resolved SMILES structures (41,240 of 102,850 CT pairs), we compute Murcko scaffolds [Bemis and Murcko, 1996] using RDKit. Pairs are grouped by scaffold, then scaffolds are assigned to train/val/test folds. The remaining 61,610 pairs without SMILES are assigned NULL scaffolds and randomly distributed. This tests whether models generalize to structurally novel drug classes.

H.5 Degree-Balanced Splitting (DTI, PPI)

Following Zheng and Wu [2020], entities are binned by their interaction degree (number of partners), and each bin is independently split into train/val/test. This ensures that high-degree and low-degree entities are proportionally represented in each fold, preventing evaluation bias toward well-studied entities. In our experiments, DDB performance was similar to random splitting across all domains (Table 2), suggesting degree imbalance is not a major confound in NegBioDB.

H.6 Control Negative Generation

For Experiment 1 (negative source inflation), we generate two types of control negatives:

- **Uniform random:** Randomly sampled entity pairs not present in the positive set or NegBioDB negatives. Equal in size to the NegBioDB negative set.
- **Degree-matched:** Random pairs where each entity's degree matches the degree distribution of the NegBioDB negative set. This controls for the hypothesis that degree alone explains performance differences.

Both control sets are generated per-seed for CT and PPI (3 seeds) and once for DTI (seed 42). Conflicts between control negatives and positive pairs are removed before training.

I Croissant Metadata

NegBioDB provides Croissant JSON-LD metadata [Akhtar et al., 2024] for machine-readable dataset discovery, as required by the NeurIPS 2026 Evaluations & Datasets Track. The metadata file (`croissant.json`) is distributed alongside the dataset on HuggingFace and describes all three domain databases and their ML export files.

The Croissant metadata includes:

- **Dataset-level fields:** Name, description, license (CC BY-SA 4.0), URL, creator, date published, citation.
- **Distribution:** Three SQLite databases (`negbiodb.db`, `negbiodb_ct.db`, `negbiodb_ppi.db`) and three Parquet ML export files.
- **Record sets:** Schema definitions for each domain's primary fact table (`negative_results`, `trial_failure_results`, `ppi_negative_results`) including field names, types, and descriptions.
- **ML-specific fields:** Split definitions, positive/negative label semantics, and recommended evaluation metrics per domain.

The metadata validates successfully with the `mlcroissant` Python library (v1.0+). A minimal excerpt:

```
{
  "@context": {"@vocab": "https://schema.org/",
               "cr": "http://mlcommons.org/croissant/"},
  "@type": "cr:Dataset",
  "name": "NegBioDB",
  "description": "Multi-domain database of experimentally confirmed
                  negative results in biomedicine",
  "license": "https://creativecommons.org/licenses/by-sa/4.0/",
  "distribution": [
    {"@type": "cr:FileObject",
     "name": "negbiodb.db",
     "contentUrl": "https://huggingface.co/.../negbiodb.db",
     "encodingFormat": "application/x-sqlite3"},
    {"@type": "cr:FileObject",
     "name": "negbiodb_ct.db", ...},
    {"@type": "cr:FileObject",
     "name": "negbiodb_ppi.db", ...}
  ],
  "recordSet": [
    {"@type": "cr:RecordSet",
     "name": "dti_negative_results",
     "field": [
       {"name": "compound_id", "@type": "cr:Field",
        "dataType": "sc:Integer"},
       {"name": "target_id", ...},
       {"name": "confidence_tier", "dataType": "sc:Text"},
       {"name": "activity_value", "dataType": "sc:Float"},
       ...
     ]}
  ]
}
```

The full Croissant JSON-LD file is available in the dataset repository.

NeurIPS Paper Checklist

1. **Claims**

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction (Section 1) state four specific contributions—NegBioDB database, NegBioBench benchmark, negative source inflation analysis, and the opacity gradient—all supported by experimental results in Section 4 with exact numerical values.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 includes a dedicated limitations paragraph addressing: solo authorship, DTI single seed, CT drug resolution coverage (20.6%), PPI L1/L2 trivial solvability, L3 judge ceiling effect, and contamination analysis scope.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This paper presents empirical results and a benchmark; it does not include theoretical results or proofs.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 describes all models, splits, and configurations. Appendix H details splitting strategies, Appendix F provides full LLM prompt templates, and Appendix G provides database schema. All hyperparameters and random seeds are specified. Complete per-run results appear in Appendices B–C.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The database (SQLite files), ML exports (Parquet), and all source code are released via GitHub and HuggingFace under CC BY-SA 4.0 (data) and MIT (code) licenses. Croissant metadata is provided for machine-readable dataset discovery. SLURM scripts for HPC execution are included.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 specifies all models, split strategies, and evaluation metrics. Appendix H provides complete split details including ratios, temporal cutoffs, and METIS parameters. LLM configurations (zero-shot and 3-shot with seeds 42/43/44) are specified.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: CT and PPI experiments use 3 random seeds (42, 43, 44); results are reported as mean $\pm$ standard deviation. DTI uses a single seed, which is acknowledged as a limitation in Section 5. LLM 3-shot results report mean $\pm$ std across 3 independent example sets.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: ML experiments were run on Cornell Cayuga HPC cluster (NVIDIA A100 GPUs, 40GB). Local LLMs (Llama, Qwen, Mistral) used vLLM on A100s. API-based LLMs used commercial endpoints (OpenAI, Google, Anthropic). Database construction ran on Apple M1 Mac (64GB RAM). Total compute: approximately 500 GPU-hours for ML training, 200 GPU-hours for local LLM inference.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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   Justification: All data is derived from public databases with appropriate licenses. No human subjects or private data are involved. The work aims to improve biomedical AI evaluation, with potential benefits for drug discovery and clinical trial design.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 5 discusses positive impacts (improving drug discovery, preventing wasted research effort) and negative impacts (100% evidence hallucination rate as a safety concern for LLM deployment in clinical settings). The opacity gradient finding directly warns against deploying LLMs for negative result assessment without contamination controls.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: NegBioDB contains only publicly available biomedical data (compound structures, protein sequences, clinical trial metadata) aggregated from 12 public databases. No pre-trained models are released. The data poses no risk for misuse beyond its source databases.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All 12 source databases are cited with their licenses: ChEMBL (CC BY-SA 3.0), PubChem (public domain), BindingDB (CC BY 3.0), DAVIS (CC BY 4.0), AACT (public domain), CTO (MIT), Open Targets (Apache 2.0), Shi & Du 2024 (CC BY 4.0), IntAct (CC BY 4.0), HuRI (CC BY 4.0), hu.MAP (public), STRING (CC BY 4.0). NegBioDB adopts CC BY-SA 4.0 to comply with ChEMBL's viral share-alike clause.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: NegBioDB is documented via: Gebru et al. Datasheet (Appendix A), database schema (Appendix G), Croissant JSON-LD metadata, HuggingFace dataset card, and comprehensive README. The benchmark code includes 800+ automated tests.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This research does not involve crowdsourcing or human subjects. All data comes from public databases and all experiments are computational.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research is involved. Clinical trial data is sourced from the public AACT database (de-identified).

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: LLMs are a core component of the benchmark evaluation (Section 3, LLM Track). Five LLMs are evaluated as subjects across 4 levels and 3 domains (241 total runs). GPT-4o-mini is additionally used as an LLM-as-judge for L3 reasoning evaluation. All model identifiers, versions, and configurations are specified. Claude Code was used for code development and paper writing assistance.